UMA SRINIVAS

Senior Software Engineer

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SUMMARY

Full Stack Engineer with 8 years of experience in designing and developing web and cloud-based applications. Skilled in analysis, architecture, development, implementation, and testing of software solutions across web, cloud, and desktop environments. Committed to driving innovation, optimizing performance, and delivering high-quality solutions that meet business objectives.

PROFESSIONAL SKILLS

Fronted Technologies : HTML5, CSS, SCSS, Boot Strap, JavaScript, TypeScript, ES8, React.js, Storybook, Recoil, Redux, Redux Saga, Redux ToolKit, Angular, Next.js, Jest, Enzyme, Gatsby, Webpack

Backend Technologies : Node.js, Express.js, Java, Spring Boot, JPA, Socket.IO, Python, JWT, OAuth, SSO, SQL, PostgreSQL, Oracle, MongoDB, NoSQL, RabbitMQ, Elastic Search

AI/ML : Rasa NLU, Microsoft Bot Framework

Tools : Elastic Search, Jira, Confluence, Azure Devops

DevOps : Docker, Dynatrace, Jenkins, Kubernetes, Camunda BPMN, NPM, Git, AWS

Operating System : Windows, Linux

WORK EXPERIENCE

Walmart Inc — Mar 2025 – Till Now

TrafficSense UI - Software Engineer III

- Led the creation of a custom data analytics dashboard to replace Adobe Analytics, enhancing data control and eliminating external licensing costs.
- Involved in cross-functional collaboration between engineering and product teams to define analytics KPIs, dashboards, and feature rollouts.
- Architected complex React workflows with advanced patterns (Context API, Redux, and Custom Hooks), enabling scalable state management and real-time UI updates across data-intensive modules.
- Developed reusable, accessible components in Storybook and managed them in shared repositories, while enhancing responsiveness with Bootstrap and media queries to ensure a seamless multi-device experience.
- Improved front-end performance and responsiveness by implementing layout effects, debouncing, memorization, lazy loading, and API response caching, while optimizing data flow for high-volume charts using efficient state management and conditional rendering.
- Integrated AG Grid and AMCharts for dynamic, enterprise-grade data visualizations and reporting features with filtering, and export functionalities.
- Integrated an in-house chatbot package into the React dashboard, resolving TypeScript module interoperability and Axios-based communication issues using Webpack overrides and custom module resolution, ensuring seamless functionality and maintainable integration.
- Built and secured RESTful APIs using Spring Boot and Java, incorporating JWT token validation and Single Sign-On (SSO) with login forms aligned with enterprise authentication standards.
- Collaborated with data engineers to design and implement complex SQL and aggregation queries, and optimized APIs to improve performance, reducing dashboard load times by 40–60% and enhancing data accuracy across business units.
- Utilized Project Lombok to streamline the creation of Data Transfer Objects (DTOs), reducing boilerplate code and improving development efficiency across backend services.
- Achieved over 90%-unit test coverage across the codebase with robust testing frameworks, including Jest, React Testing Library, JUnit, and JaCoCo.
- Handled CI/CD pipelines, ensuring smooth delivery cycles and minimizing deployment issues by following Agile workflows such as Scrum and Kanban.methodologies.

NRG Energy — Sep 2023 – Feb 2025

Direct Energy Texas - Programmer Analyst III

- Delivered mobile-responsive & accessible web application for enrollment and account management of customer's electricity subscriptions by aligning with cross-functional and cross-brand requirements.
- Led the design and development in Agile methodologies enabling users to manage profiles, electricity plans, TOS & EFL, utilization, and payments
- Architected reusable, scalable UI components using React.js, TypeScript, and React Hooks, streamlining development and consistency across applications.
- Ensured web accessibility compliance (WCAG) across all pages using storybook add-on-ally with Internalization support using i18next.
- Introduced Formik with Yup for robust form validation, improving user input handling efficiency.
- Developed reverse proxies using Node.js & Express.js and created Restful Web APIs using Java, Spring boot to access data from 3rd party services.
- Leveraged Redux Toolkit and AWS Cognito for secure authentication and efficient REST API data fetching.
- Troubleshooted critical cross-platform issues, maintaining a consistent user experience across devices.
- Incorporated Adobe Analytics to track user behavior, enabling data-driven UX optimizations.
- Achieved front-end responsiveness using SCSS, Bootstrap, and media queries, delivering multi-device experience.
- Added secure server side [SSR] JSP pages in Spring with JPA for agent-driven enrollments and payment processing, ensuring PCA compliance.
- Improved unit test coverage to over 80% using Jest and React Testing Library, ensuring reliability of application.
- Automated CI/CD pipeline with Jenkins, maintaining zero vulnerabilities with SonarQube and reducing deployment errors by 30% by following scrum or kanban methodologies.

Ivy Software Development Services Pvt Ltd

Apr 2021 – Apr 2023

Omnia - Software Engineer

- Developed a high-performance back-office gambling application using, React.js, Electron.js, and a Micro Frontend architecture, enabling shop colleagues to manage bets, process payouts, and generate slips efficiently.
- Implemented concurrent transaction workflows using Redux Sagas to coordinate external betting service calls and internal backend updates with transaction IDs, and integrated Electron-based print services to automatically generate betting slips upon confirmation
- Embedded third-party betting interfaces via iframes and implemented event listener communication to capture bet events, store transaction details in the backend, and redirect users to payout workflows.
- Planned and implemented real-time notifications using Socket.IO for instant updates on bet transactions and terminal events, improving operational transparency and response times.
- Migrated legacy Angular modules and refactored class-based React components to reusable functional components, standardizing central artifacts for maintainability and scalability.
- Achieved 90% test coverage through Jest unit and integration tests and automated deployment workflows using AWS, Jenkins, and CI/CD pipelines, enhancing system reliability, accelerating delivery cycles, and ensuring compliance and operational integrity.

Digitamize Innovation pvt ltd

Feb 2019 – Apr 2021

Chatbot Low-code Platform - Product Engineer

- Led and Managed the team in development of a low-code Chatbot Platform, delivering customizable NLP-based solutions with Camunda BPMN for conversational flow automation.
- Built secure, scalable REST APIs using Node.js and Express.js, adhering to OWASP guidelines and optimizing performance for microservices architecture ensuring scalability & performance.
- Implemented robust authentication and authorization mechanisms for REST APIs by integrating LDAP and OAuth with Keycloak SSO, enabling CORS and enforcing security policies to safeguard each request
- Made Chatbot product using Microsoft Bot Framework and Improved its usability by integrating Elasticsearch for auto-suggestions and related queries.
- Led and Conducted knowledge transfer sessions on NLP training and Camunda BPMN dialogue flows to ensure team alignment and collaboration.
- Engineered Event driven and Web Socket mechanisms to optimize live data synchronization for critical operations.
- Enhanced the NLP Trainer dashboard using python and RabbitMQ to manage queuing mechanisms and spawn containers for training live logs.
- Use PM2 and cluster modules of Nodejs and efficiently Utilized worker threads to offload CPU-intensive tasks, improving main thread responsiveness.
- Developed and optimized Webpack configurations to reduce bundle sizes, and regularly utilized Linux CLI and shell scripts for automation, environment setup, and build/deployment processes, ensuring seamless SaaS release compatibility.
- Dockerized 9 microservices and configured NGINX as a reverse proxy with SSL termination and security headers to fortify application security, and resolved vulnerabilities identified by the Nessus scanner

Fortunapix Pvt Ltd

Jan 2018 – Apr 2019

LernBook - Associate Software Engineer

- Contributed to innovative client solutions by developing proof-of-concept and documenting findings in using Ontologies for Book resources.
- Assisted & Collaborated with the team to utilize JavaScript tools for extracting PDF content and generating digital books with Apache POI library.
- Implemented REST APIs and core services from business process using Java, Spring boot, and MongoDB, enabling CRUD operations for digital book managent and user controls
- Automated content ingestion to store hierarchical book data in MongoDB and assets in AWS cloud S3, reducing manual processing time by 90%.
- Assisted developers by revising JSON structures and RESTful APIs to provision precise data based on requirements.
- Deployed a modular, Kubernetes-based architecture, enabling resource allocation and cost efficiency.
- Incorporated offline book access into a mobile app built with Ionic and Angular, preloading books for remote access.

EDUCATION

Bachelors of Technology – Electrical and Electronics Engineering, Andhra University, 2017

CERTIFICATIONS

- **AWS Certified Developer Associate**
- **Apollo Certified GraphQL Developer**

Blogs & Contributions

GitHub

https://github.com/kattulaumasrinivas

Blog

https://umasrinivask.medium.com/dockerize-elasticsearch-kibana-with-x-packs-security-237589acb3fd

Blog

https://umasrinivask.medium.com/react- application-deployments-with-runtime- configuration-dd42820e3f50

Blog

https://umasrinivask.medium.com/configure-nginx-as-a-reverse-proxy-with-docker-compose- file-4ebba2b75c89

Blog

https://umasrinivask.medium.com/build-a-multi-stage-docker-image-for-react-app-to-load-runtime-environment-settings-911517f5f58d

Registry

https://hub.docker.com/u/kattulaumasrinivas